

Complex Systems Seminar: Week 1 — Networks and Emergence

1. Foundations of Agent-Based Modeling (ABM)

Agent-based modeling (ABM) represents an ontological commitment to viewing reality through the lens of an "abstracted description." Far from being a mere technical simulation, ABM demands a strategic simplification of reality to make complex phenomena tractable. We begin our inquiry by embracing the "useful wrongness" of models, a concept popularized by statistician George Box. In the architecture of complex systems, a model that sought a one-to-one correspondence with reality would be as unusable as the world it mimics. This is best illustrated by Jorge Luis Borges' "Del rigor en la ciencia," where geographers construct a map so detailed it occupies the same space as the empire itself—a representation rendered futile by its own lack of abstraction.

To facilitate this scientific simplification, we utilize the NetLogo environment, structured around three primary entities. Their nomenclature reflects the field's historical lineage:

Entity	Role in NetLogo	Function
Turtles	Movable Agents	Primary actors. The term originates from Seymour Papert (1969) and the Logo language, named after physical robot "turtles" used to teach children computational logic.
Patches	Environmental Background	The immobile, discrete grid units (squares) that define the landscape or spatial topology inhabited by agents.
Links	Relationship Mapping	The explicit ties defining social networks or structural dependencies between agents.

John Holland famously compared models to "political cartoons." By exaggerating salient features—the "nose" of a policy or the "eyes" of a behavioral rule—while backgrounding irrelevant noise, researchers isolate the causal mechanisms that drive system behavior. Once we accept this epistemic humility and the necessity of abstraction, we can examine how these discrete agents produce the macro-patterns of emergence.

2. The Mechanics of Emergence and Feedback Loops

The strategic core of computational social science is the gap between individual-level rules and aggregate-level patterns. In complex systems, the "whole is more than the sum of its parts" because of nonlinear interactions. Emergence is the process where micro-level behaviors coalesce into macro-structures that no individual agent explicitly controls. Consider the "Traffic Basic" model. Simple rules—accelerating when clear, braking when an agent is ahead—do not just create slow movement; they generate "Stop-and-Go" waves. Standard statistical metrics like "mean speed" are fundamentally deceptive here; in a system where agents oscillate between stasis and maximum velocity, the average captures a state that almost no agent ever actually inhabits. Furthermore, the jam itself emerges as a distinct entity that traverses the system backward—a macro-pattern moving in the opposite direction of the micro-agents.

This logic extends to social emergence and the reduction of "social friction." A seminal historical example is the 1975 "Slugging" phenomenon on Washington DC's Shirley Highway. When top-down High Occupancy Vehicle (HOV) lanes were instituted, the individual desire for speed (a micro-goal) incentivized strangers to form a bottom-up social

system. To mitigate the risk and friction of carpooling with strangers, a complex set of emergent social norms crystallized:

- The driver maintains absolute radio and window control.
- Prohibitions on smoking, eating, and passenger makeup application.
- Specific safety protocols, such as never leaving a woman alone in the slugging queue.

These rules were never legislated; they emerged as individuals negotiated constraints to maximize their own velocity. Studying these phenomena requires a distinct scientific methodology.

3. The Generative Turn: ABM as the "Third Way" of Science

Positioning ABM within the history of scientific inquiry, Robert Axelrod (1997) defined simulation as a "third way" of doing science. It bridges the gap between the data-driven inferences of induction and the first-principles reasoning of deduction.

Mode	Methodology	Primary Goal
Induction	Inference from data	Synthesizing theories from observed patterns.
Deduction	Reasoning from first principles	Deriving logical consequences from axioms.
Generation	Growing data from principles	Using simple rules to "grow" aggregate patterns.

This generative approach enables two types of theoretical insight. **Integrative Understanding** allows us to start with micro-rules (e.g., Flocking: separate, align, cohere) and observe how they synthesize into a macro-pattern. Conversely, **Differentiated Understanding** allows us to take an observed macro-pattern (a bird flock) and work backward to hypothesize the underlying individual rules. Crucially, even if we know the rules of flocking, the human mind cannot naturally predict the resulting swarming dynamics without generative aid. ABM allows us to "grow" the data to see if our hypothesized micro-rules actually produce the empirical reality.

4. Strategic Indicators for Deploying ABM

ABM is a specialized instrument, most effective when traditional equation-based models (EBM) reach their limits. There are six primary indicators for its deployment:

1. **Medium Numbers:** Identified by John Casti (1996) via the "El Farol Bar" problem, ABM excels where populations are too large for simple game theory but too small or heterogeneous for the "mean field" averages of statistical physics.
2. **Heterogeneity:** When individual variance (wealth, thresholds, age) drives system outcomes.
3. **Complex but Local Interactions:** When agents lack a "global controller" and rely on local information.
4. **Rich Environments:** When spatial (GIS) or network topologies are fundamental to decision-making.
5. **Time (Necessary Condition):** ABM is a model of processes and temporal iterations ("ticks").

6. **Adaptation (Sufficient Condition):** If agents change behavior based on history-contingent experience, ABM is essential.

The advantage of ABM's "natural ontology" is best illustrated by the "Dwarf Wolf" problem. EBMs treat populations as continuous variables, theoretically allowing for "a millionth of a wolf" to survive and rebound. In an ABM, wolves are discrete agents; if the last pair dies, the population is extinct. This indivisibility mirrors the biological and social reality EBMs often obscure.

5. Case Study: The "Heroes and Cowards" Model

Originating from the Fratelli Theatre Group's exercises in the 1980s, this model demonstrates how slight perturbations in logic lead to diametrically opposed emergent states.

Logic and Technical Implementation: Agents (Turtles) are randomly assigned a "friend" and an "enemy" (one-of other turtles). The behavior is dictated by two phases:

- **Heroes Phase:** Agents attempt to place themselves *between* the friend and enemy. The technical midpoint is calculated as: $\text{facexy} ([\text{xcor}] \text{ of friend} + [\text{xcor}] \text{ of enemy}) / 2$ $([\text{ycor}] \text{ of friend} + [\text{ycor}] \text{ of enemy}) / 2$.
- **Cowards Phase:** Agents attempt to place the *friend* between themselves and the enemy. This requires projecting a point on the "far side" of the friend: $\text{facexy } [\text{xcor}] \text{ of friend} + ([\text{xcor}] \text{ of friend} - [\text{xcor}] \text{ of enemy})$ $[\text{ycor}] \text{ of friend} + ([\text{ycor}] \text{ of friend} - [\text{ycor}] \text{ of enemy})$.

Emergent Outcomes: The "Heroes" logic leads to a central collapse, forming a tightly packed "ball" or cluster. The "Cowards" logic leads to system-wide expansion; if the environment is bounded, agents exhibit "border-hugging" behavior. Depending on the random seed and initial distribution, a full taxonomy of patterns emerges: **dot**, **frozen**, **spiral**, **seductive**, **yo-yo**, or the **deambulatory flock**. Essential commands for this implementation include `setxy`, `facexy`, `fd`, and the critical `reset-ticks`.

6. The Role of Randomness and Documentation

The strategic necessity of controlling stochastic perturbations is vital for "scientific diplomacy." Because computers rely on deterministic algorithms to simulate chance, we utilize pseudo-randomness. By fixing a **Random Seed** (e.g., the "preset" buttons in El Farol Bar simulations), a researcher can ensure replicability, allowing others to witness the exact same "seductive" or "frozen" patterns.

Scientific validity also demands rigorous documentation via the NetLogo **Info Tab**. To prevent the conceptual model from drifting away from the implemented code, documentation must be **simultaneous** with coding. A standard checklist includes:

- **What is it?** High-level system description.
- **How it works:** Explanation of agent-level rules (Pseudocode).
- **Things to notice:** Key emergent patterns and transitions.
- **Extending the model:** Suggestions for future research.

These foundational principles prepare the researcher to transition from "cartoon" models like Heroes and Cowards to empirical validation, such as the Aché foraging model. This next step utilizes quantitative anthropological data to map individual hunters' decisions against

real-world caloric returns, moving the "Third Way" from theoretical exploration to empirical science.